INTERVIEW CHEAT SHEET

[SELF INTRODUCTION]

Nikhil Gupta, B.Tech in Computer Science (AI & ML) from Axis Institute, proficient in Python, ML, DL, Power BI, and generative AI, aiming to build intelligent solutions that drive business insights.

[PROJECT PITCHES]

-> HR Analysis Dashboard (Power BI)

Built interactive charts and KPIs from employee data; faced large dataset handling and real-time refresh challenges, overcame by using DAX measures and incremental data load.

-> Object Detection (Core Project)

Developed a basic object detection pipeline with OpenCV and ML models; struggled with limited data and accuracy, solved through data augmentation and hyperparameter tuning.

-> Google Translator with GUI

Created a desktop translator using Python GUI and APIs; latency and translation accuracy were issues, addressed by caching results and selecting reliable API endpoints.

[KEY TECHNOLOGIES]

Python	versatile language for data science, AI, and automation.
Power BI	business analytics tool for interactive data visualization.
OpenCV	computer vision library for image and video processing.
Generative AI LLMs	large language models for content generation.
Chroma/Pinecone	vector databases for similarity search and retrieval.
Git	distributed version control system for code collaboration.
Jupyter	interactive notebook environment for data exploration.
CSS/HTML	front-end web styling and markup languages.

[DSA TOPICS TO REVISE]

• **Arrays & Strings**	fundamental data structures for efficient manipulation.
• **Linked Lists**	dynamic, ordered data structure with constant-time insertions.
• **Trees & Graphs**	hierarchical and network data representations for complex queries.
• **Dynamic Programming**	optimization technique for overlapping subproblems.
• **Sorting & Searching**	core algorithms for data organization and retrieval.
• **Hash Tables**	constant-time lookup structure for key-value pairs.
• **Recursion**	solving problems by self-referential function calls.

[HR ANSWERS]

Q: Why this role	I am passionate about applying AI/ML to solve real-world business problems, and this role offers the opportunity to work on data-driven solutions that impact decision-making.
Q: Strength	Strong problem-solving skills, end-to-end ML pipeline experience, and the ability to translate technical insights into actionable business outcomes.
Q: Weakness	I tend to over-commit to multiple projects; I am learning to prioritize tasks and delegate effectively.
Q: Long term goal	To lead an AI research team that develops innovative solutions for industry challenges and drives digital transformation.
Q: Why you hired in this company	Your company's focus on cutting-edge AI aligns with my expertise, and I can contribute to your product roadmap by delivering data-driven insights and scalable ML solutions.
Q: What is your salary expectation	Based on market rates and my experience, I expect a package in the range of ?12-15 LPA.

[WEAK AREAS TO COVER]

>> Deep learning model deployment on cloud platforms

>> Advanced cloud architecture and services

>> Advanced statistical modeling techniques

>> Negotiation and stakeholder management skills